

typst:

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$
&not (p and (q or r)) \
&= not p or (not q and not r) & "(demorgan twice)" \
&= (not p or not q) and (not p or not r) wide &
"(distributive)" \
&= not (p and q) and not(p and r) & "(demorgan)"
$

```

latex:

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\begin{aligned}
& \& \neg (p \& (q \vee r)) \\\
& = \neg p \vee (\neg q \wedge \neg r) \& \text{(demorgan twice)} \\\
& = (\neg p \vee \neg q) \wedge (\neg p \vee \neg r) \& \text{(distributive)} \\\
& = \neg(p \wedge q) \wedge \neg(p \wedge r) \& \text{(demorgan)}
\end{aligned}

```

$$\begin{aligned}
 & \neg(p \wedge (q \vee r)) \\
 & = \neg p \vee (\neg q \wedge \neg r) && \text{(demorgan twice)} \\
 & = (\neg p \vee \neg q) \wedge (\neg p \vee \neg r) && \text{(distributive)} \\
 & = \neg(p \wedge q) \wedge \neg(p \wedge r) && \text{(demorgan)}
 \end{aligned}$$